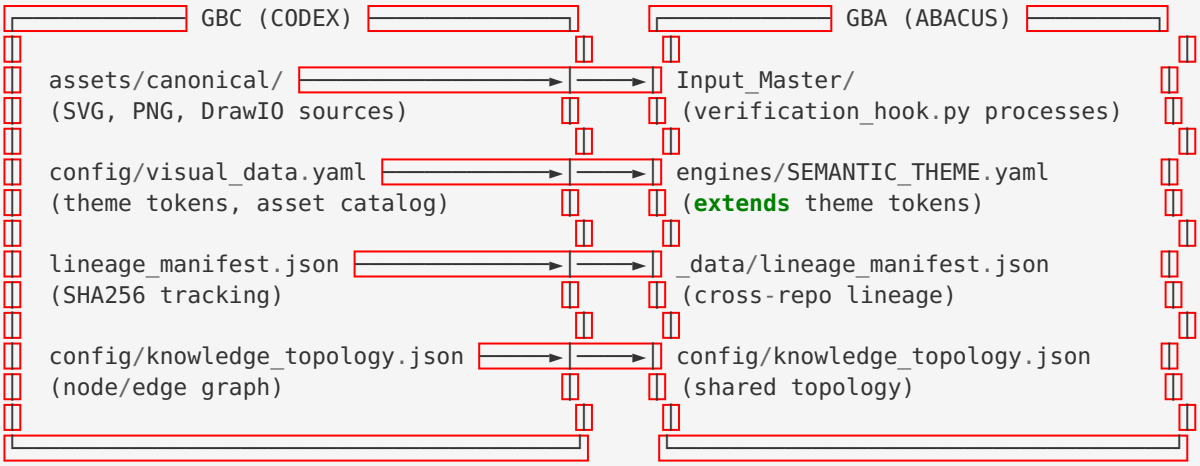
GBA Integration Guide

> How GBOGEB/CODEX (GBC) integrates with GBOGEB/ABACUS (GBA)

Overview

GBC serves as the **visual asset governance layer** — the canonical source for engineering diagrams, schematics, and SVG assets. GBA is the **publication governance engine** that consumes these assets for rendering in dashboards, reports, and presentations.



Integration Points

1. Asset Transfer Pipeline

```
# 1. Export compiled SVGs from GBC
cd gbogeb_codex
python src/svg_cleaner.py --input-dir assets/canonical/ --output-dir assets/compiled/

# 2. Copy to GBA Input_Master
cp assets/compiled/*.svg ../gbogeb_abacus/Input_Master/

# 3. Run GBA verification hook
cd ../gbogeb_abacus
python engines/verification_hook.py --input-dir Input_Master/ --data-dir _data/
```

2. Knowledge Topology

Both repositories maintain `config/knowledge_topology.json` files that reference each other:

GBC Node	Relation	GBA Node
`GBC-ASSET-001` (cryomodule.svg)	`consumed_by`	`GBA-HMI-001` (HMI Dashboard)
`GBC-ASSET-002` (facility_layout.svg)	`referenced_in`	`GBA-REPORT-001` (Tech Report)
`GBC-ENGINE-001` (asset_validator.py)	`mirrors`	`GBA-ENGINE-HOOK` (verification_hook.py)
`GBC-ENGINE-003` (lineage_tracker.py)	`feeds`	`GBA-LINEAGE-001` (lineage_manifest.json)
`GBC-CONFIG-001` (visual_data.yaml)	`extends`	`GBA-THEME-001` (SEMANTIC_THEME.yaml)

3. Theme Token Mapping

GBC's `visual_data.yaml` defines diagram-specific theme tokens that extend GBA's semantic theme:

```yaml
# GBC visual_data.yaml
theme_tokens:
  diagram_stroke: "var(--color-primary)"      # → GBA SEMANTIC_THEME primary
  diagram_fill_surface: "var(--color-surface)" # → GBA SEMANTIC_THEME surface
  status_nominal: "var(--color-success)"       # → GBA SEMANTIC_THEME success
```

4. Lineage Continuity

GBC's `lineage_manifest.json` and GBA's `_data/lineage_manifest.json` form a **cross-repository lineage chain**:

```
GBC: canonical SVG → sha256 → GBC manifest
                                ↓
GBA: Input_Master/ → verification_hook → GBA manifest → rendered output
```

5. Archive Bundle

GBC generates `gbc_asset_bundle.tar.gz` containing:
- Compiled/optimized SVGs
- Visual data registry
- Knowledge topology
- Stakeholder registry
- Lineage manifest

This bundle can be consumed directly by GBA's CI/CD pipeline.

Validation Checklist

- [ ] All SVGs in `assets/canonical/` pass `asset_validator.py`
- [ ] `lineage_manifest.json` is valid (`lineage_tracker.py --validate`)

- [ ] SHA256 hashes match on-disk files (`lineage_tracker.py --verify-hashes`)
- [ ] `knowledge_topology.json` nodes have matching GBA references
- [ ] Theme tokens in `visual_data.yaml` map to GBA `SEMANTIC_THEME.yaml`
- [ ] CI/CD pipelines complete without errors